

# **Lecture 30**

**STAT 109: Introductory Biostatistics**

# Lecture 30

## Four Complete Examples for Causal Questions and Two Group Comparisons

This lecture walks through four complete examples that reuse descriptive statistics and plots from Lectures 25–27 and inference tools from Lectures 28–29.

### Learning objectives

Use the **Methods table** below to pick a **test** for the “by chance” explanation:

- For two groups with a numerical outcome, produce Lecture 25 summaries (`group_by`, `summarize`, boxplots, optional histograms) and run a Welch `t.test()` after checking conditions.
- For two categorical variables, produce Lecture 27 tables and stacked bar charts, then run `prop.test` if binary or `chisq.test()` after inspecting expected counts.

Write a short conclusion that names study design (RCT vs observational) and whether causal language is appropriate.

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## Choosing a test for “by chance” (Methods table)

Use this table when you compare **exactly two groups** defined by a **binary explanatory** variable (treatment A vs B, university 1 vs 2, exposed vs not). It matches and the tools from **Lectures 25, 27, 28, and 29**.

| Explanatory (two groups) | Response type | Population summary<br>you target (aka<br>parameter of<br>interest) | Procedure                                                                                                                                                                                                        | Conditions / checks                                                                                                                                                                            |
|--------------------------|---------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Binary                   | Numerical     | Difference in means<br>$\mu_1 - \mu_2$                             | <b>Welch two-sample<br/><i>t</i>-test:</b> <code>t.test(y ~<br/>group, data =<br/>..., var.equal =<br/>FALSE)</code> ; optional<br><b>Rossman<br/>randomization<br/>(two groups)</b> on a<br>difference in means | <b>Independent</b><br>observations (or<br>random assignment<br>AND large enough<br>group sizes (both<br>groups are larger<br>than 20) or roughly<br>symmetric<br>populations in each<br>group. |

| Explanatory (two groups) | Response type                            | Population summary you target (aka parameter of interest)                    | Procedure                                                                                                                                                 | Conditions / checks                                                                                                                                                                                     |
|--------------------------|------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Binary                   | <b>Binary</b> (or two-level categorical) | Difference in <b>proportions</b> $p_1 - p_2$ (or association in a 2×2 table) | <b>prop.test()</b> on counts from a binary comparison;<br><b>chisq.test()</b> on a <b>contingency table</b> of counts; optional <b>Chi-square shuffle</b> | <b>Independent</b> observations AND for Pearson <b>chi-squared</b> , require <b>all expected counts</b> ≥ 5 (inspect <code>chisq.test(...)\$expected</code> ); otherwise <b>simulate.p.value = TRUE</b> |

Note: this table doesn't suggest methods for comparing more than two groups, paired measurements, or numerical × numerical relationships.

Hypotheses reminder. For means,  $H_0: \mu_1 = \mu_2$ ; for proportions in a 2×2 table,  $H_0: p_1 = p_2$ . Always define symbols in words, e.g.  $\mu_1$  is the mean of the response variable in group 1,  $p_1$  is the rate of successes in the response variable for group 1 etc. IN THE POPULATION, not the sample we have collected data from.

## Example 1:

Context: adults visiting a university hospital in January-March of 2020 with a particular chronic illness are randomly assigned to Standard care or New therapy. The response is years lived after enrollment.

```
if (!requireNamespace("tidyverse", quietly = TRUE)) {
  install.packages("tidyverse", repos = "https://cloud.r-project.org")
}

library(tidyverse)

set.seed(3001)
life_rct <- tibble(
  arm = factor(
    rep(c("New therapy", "Standard care"), each = 45),
    levels = c("Standard care", "New therapy")
  ),
  years_after_enroll = c(
    rnorm(45, mean = 6.4, sd = 1.05),
    rnorm(45, mean = 5.0, sd = 0.95)
  )
)
```

## Descriptive statistics (Lecture 25)

```
life_rct |>
  group_by(arm) |>
  summarize(
```

```

n = n(),
mean_years = mean(years_after_enroll),
sd_years = sd(years_after_enroll),
median_years = median(years_after_enroll),
iqr_years = IQR(years_after_enroll)
)

```

# A tibble: 2 x 6

| arm             | n     | mean_years | sd_years | median_years | iqr_years |
|-----------------|-------|------------|----------|--------------|-----------|
| <fct>           | <int> | <dbl>      | <dbl>    | <dbl>        | <dbl>     |
| 1 Standard care | 45    | 4.92       | 1.04     | 4.83         | 1.27      |
| 2 New therapy   | 45    | 6.19       | 1.18     | 6.12         | 1.54      |

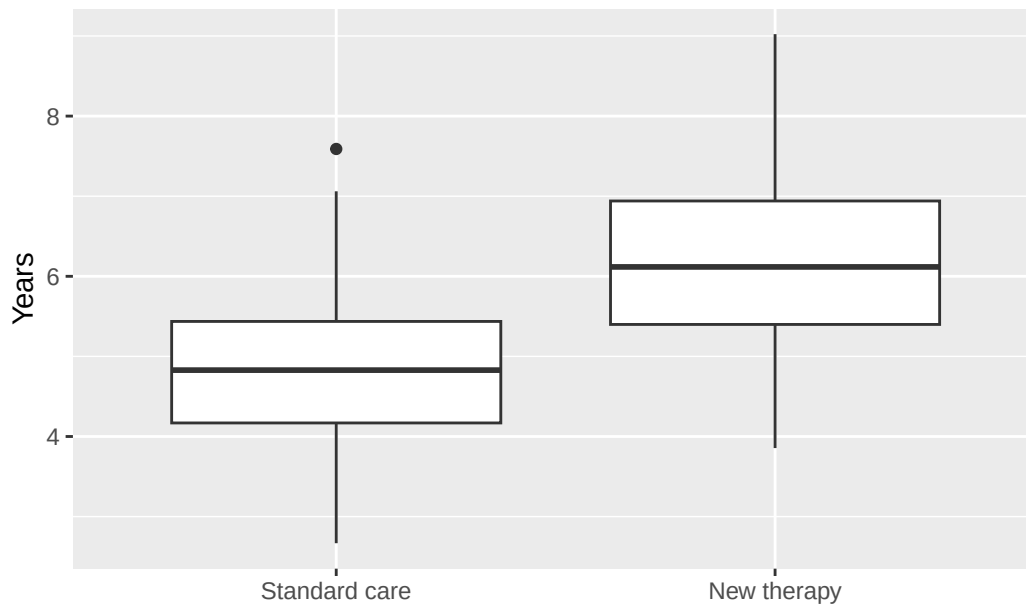
## Bivariate plots

```

ggplot(life_rct, aes(x = arm, y = years_after_enroll)) +
  geom_boxplot() +
  labs(
    title = "Years lived after enrollment by treatment arm (simulated RCT)",
    x = NULL,
    y = "Years"
  )

```

Years lived after enrollment by treatment arm (simulated RCT)

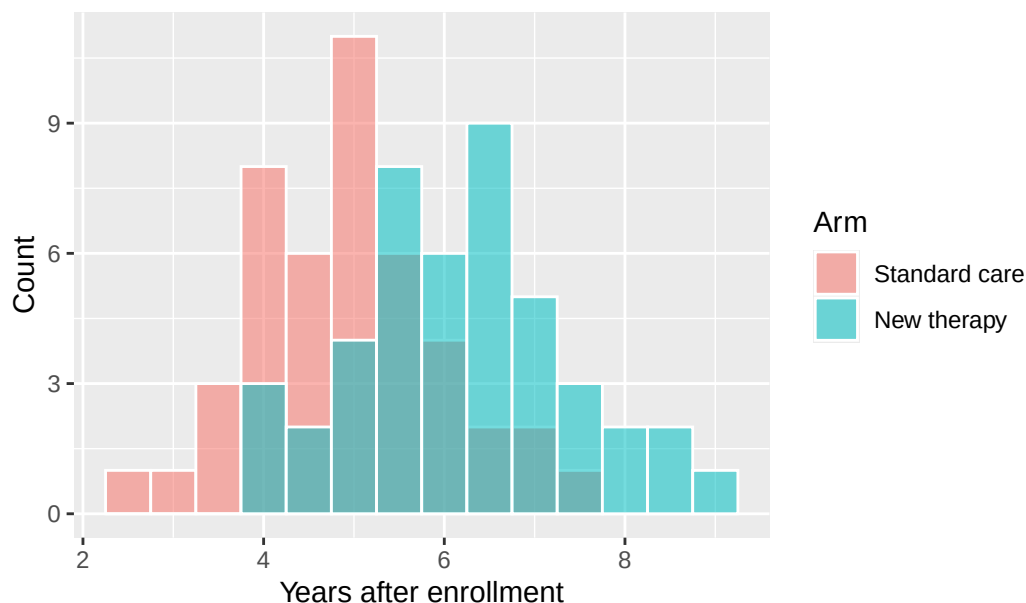


```

ggplot(life_rct, aes(x = years_after_enroll, fill = arm)) +
  geom_histogram(binwidth = 0.5, position = "identity", alpha = 0.55, color = "white") +
  labs(
    title = "Distribution of survival years by arm",
    x = "Years after enrollment",
    y = "Count",
    fill = "Arm"
  )

```

Distribution of survival years by arm



## Welch two-sample *t*-test

First, we'll check the conditions: the patients in the two groups are unlikely to influence one another's life span (independence) AND both groups are larger than 20.

```
t.test(years_after_enroll ~ arm, data = life_rct, var.equal = FALSE)
```

Welch Two Sample t-test

```
data: years_after_enroll by arm
t = -5.4028, df = 86.734, p-value = 5.676e-07
alternative hypothesis: true difference in means between group Standard care and group New therapy is not e
95 percent confidence interval:
 -1.7345006 -0.8014981
sample estimates:
mean in group Standard care    mean in group New therapy
          4.923762              6.191761
```

**Conclusion.** A small  $p$ -value supports the idea that the difference in mean years is not well explained by chance alone. This is an RCT, so causal language about the effect of assignment to therapy is reasonable for trial participants since the possibility of confounding was reduced by randomly assigning the participants to treatments. We cannot safely generalize from this study to the larger population of people with this chronic illness because the trial participants were non-randomly selected from only one hospital.

## Example 2

Content: Students in a particular statistics class at Cal Poly Humboldt may enroll in an optional supplemental instruction (SI) workshop. The outcome of interest is the final exam score. Prior preparation may influences both enrollment in SI and exam score.

```

set.seed(3002)
n <- 100
prior_prep <- sample(c("Low", "High"), n, replace = TRUE, prob = c(0.55, 0.45))
p_take <- ifelse(prior_prep == "High", 0.58, 0.22)
take_enrich <- rbinom(n, 1, prob = p_take)
exam_score <- 52 +
  17 * (prior_prep == "High") +
  2.2 * take_enrich +
  rnorm(n, 0, 6)

enrich_obs <- tibble(
  prior_prep = factor(prior_prep, levels = c("Low", "High")),
  program = factor(ifelse(take_enrich == 1, "Enrichment", "Standard")),
  exam_score = exam_score
)

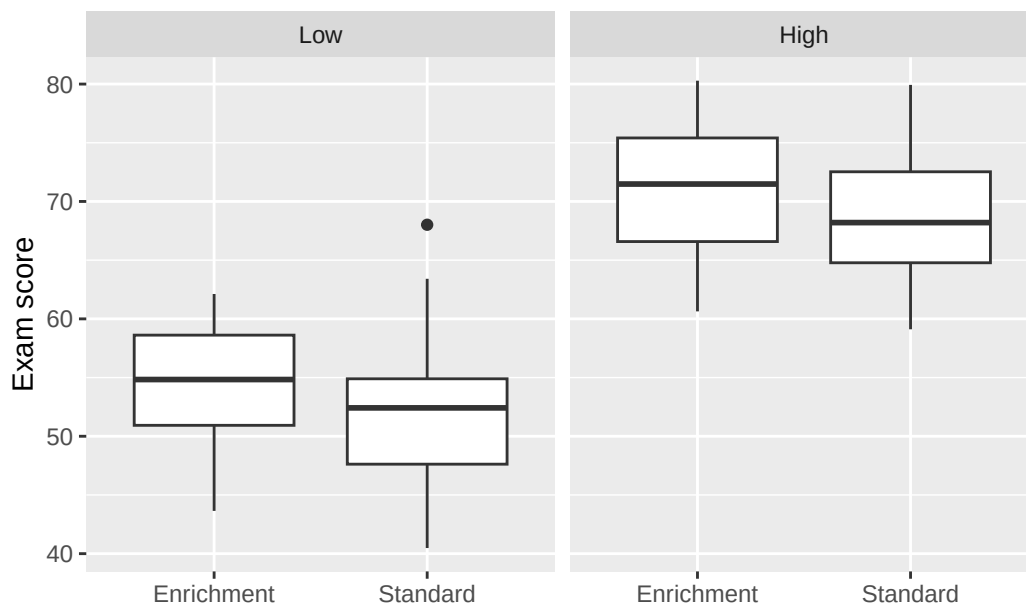
```

## Boxplots by Group

```

ggplot(enrich_obs, aes(x = program, y = exam_score)) +
  geom_boxplot() +
  facet_wrap(~ prior_prep) +
  labs(
    title = "",
    x = NULL,
    y = "Exam score"
  )

```



## Incorrect two-sample *t*-test (ignores prior preparation)

```

t.test(exam_score ~ program, data = enrich_obs, var.equal = FALSE)

```

### Welch Two Sample t-test

```
data: exam_score by program
t = 4.1863, df = 89.522, p-value = 6.617e-05
alternative hypothesis: true difference in means between group Enrichment and group Standard is not equal to 0
95 percent confidence interval:
 4.416525 12.395566
sample estimates:
mean in group Enrichment    mean in group Standard
          65.89883              57.49279
```

### Means within prior preparation (subclassification)

```
enrich_obs |>
  group_by(prior_prep, program) |>
  summarize(mean_exam = mean(exam_score), n = n(), .groups = "drop")
```

```
# A tibble: 4 x 4
  prior_prep program    mean_exam     n
  <fct>      <fct>      <dbl> <int>
1 Low       Enrichment      54.4     13
2 Low       Standard      51.8     39
3 High      Enrichment      71.1     29
4 High      Standard      69.1     19
```

**Conclusion.** A naive *t*-test may show a “significant” difference between Enrichment and Standard mean scores. Because enrollment is observational, confounding by prior preparation is a serious alternative explanation. Do not claim the workshop caused higher scores without a design that rules out confounding.

---

## Example 3

Context: Patients at a research center were randomized to Vaccine or Placebo; infection yes/no during the next follow-up was recorded.

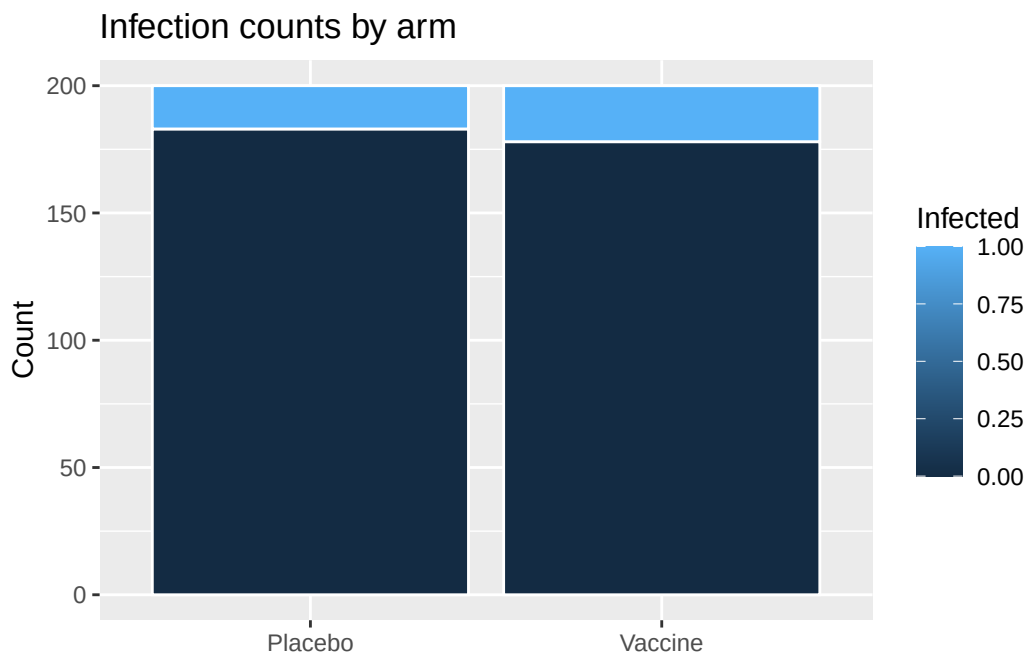
```
set.seed(3003)
vax_trial <- tibble(
  arm = factor(rep(c("Vaccine", "Placebo"), each = 200)),
  infected = c(rbinom(200, 1, 0.095), rbinom(200, 1, 0.102))
)

tab_vax <- table(arm = vax_trial$arm, infected = vax_trial$infected)
tab_vax
```

|         | infected |    |
|---------|----------|----|
| arm     | 0        | 1  |
| Placebo | 183      | 17 |
| Vaccine | 178      | 22 |

## Stacked counts

```
vax_trial |>
  count(arm, infected) |>
  ggplot(aes(x = arm, y = n, fill = infected)) +
  geom_col(color = "white") +
  labs(
    title = "Infection counts by arm",
    x = NULL,
    y = "Count",
    fill = "Infected"
  )
```



## Chi-squared test of independence — check expected counts

```
cq <- chisq.test(tab_vax)
cq$expected
```

```
      infected
arm          0    1
Placebo 180.5 19.5
Vaccine 180.5 19.5
```

```
cq
```

Pearson's Chi-squared test with Yates' continuity correction

```
data:  tab_vax
X-squared = 0.45458, df = 1, p-value = 0.5002
```

You could also run `prop.test()` on the **infected counts** by arm (Methods table row for **binary × binary**). Conclusion. With a large *p*-value, this data are consistent with the “by chance explanation: the underlying infections



rates under the two treatments are the same and our observed difference could be due to sampling a subset the populations. Because group assignment is random, the risk of confounding is reduced.

---

## Example 4

```
uni <- tibble(  
  university = rep(c("Timber Ridge", "Wild Horse"), c(200, 200)),  
  passed = c(rep(c(TRUE, FALSE), c(150, 50)), rep(c(TRUE, FALSE), c(154, 46)))  
)  
  
tab_u <- table(uni$university, uni$passed)  
chisq.test(tab_u)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data:  tab_u  
X-squared = 0.12336, df = 1, p-value = 0.7254
```

Crude pass rates: Timber Ridge 75%, Wild Horse 77%. Note that the p-value from just university and passed is insufficient for a fair comparison of the two universities because it doesn't take into account the confound of AP Calculus.

Subclassification by AP Calculus:

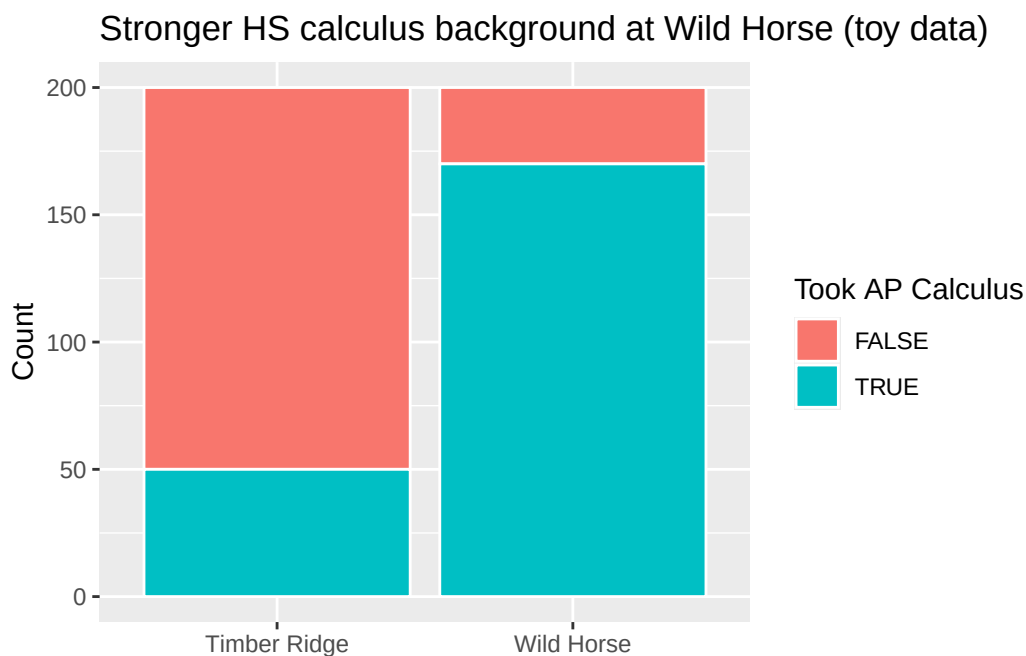
```
uni_ap <- tibble(  
  university = c(rep("Timber Ridge", 200), rep("Wild Horse", 200)),  
  ap_calc = c(  
    rep(c(TRUE, FALSE), c(50, 150)),  
    rep(c(TRUE, FALSE), c(170, 30))  
  ),  
  passed = c(  
    rep(c(TRUE, FALSE), c(45, 5)),  
    rep(c(TRUE, FALSE), c(105, 45)),  
    rep(c(TRUE, FALSE), c(136, 34)),  
    rep(c(TRUE, FALSE), c(18, 12))  
  )  
)  
  
uni_ap |>  
  group_by(university, ap_calc) |>  
  summarize(  
    pass_rate = mean(passed),  
    n = n(),  
    .groups = "drop"  
  )
```

```
# A tibble: 4 x 4  
  university ap_calc pass_rate    n  
  <chr>      <lgl>      <dbl> <int>  
1 Timber Ridge FALSE      0.7   150  
2 Timber Ridge TRUE       0.9    50
```

|   |            |       |     |     |
|---|------------|-------|-----|-----|
| 3 | Wild Horse | FALSE | 0.6 | 30  |
| 4 | Wild Horse | TRUE  | 0.8 | 170 |

### Stacked bar chart for AP mix

```
uni_ap |>
  count(university, ap_calc) |>
  ggplot(aes(x = university, y = n, fill = ap_calc)) +
  geom_col(color = "white") +
  labs(
    title = "Stronger HS calculus background at Wild Horse (toy data)",
    x = NULL,
    y = "Count",
    fill = "Took AP Calculus"
  )
```



Conclusion template. A chi-squared test on  $\text{university} \times \text{pass}$  doesn't help us decide whether a university causes the difference in passage rates. Here, AP calculus is a confounder: students with AP Calculus are more likely to attend Wild Horse AND are more likely to pass calculus. within AP groups, Timber Ridge looks better however this adjusted comparison is still not fair as there could be other unmeasured confounders, just as student motivation.